

A Pedagogical Review of The Lane-Emden Equation

A pedagogical review and worked derivation of the Lane-Emden equation, including an independent variational rederivation and a numerical comparison to the Model S solar model

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Contents

1 Derivation of the Lane-Emden from First Principles	3
1.1 Assumptions	3
1.2 Hydrostatic Equilibrium	4
1.3 Polytropic Equation of State	5
1.4 Lane Emden Equation	6
1.5 Boundary Conditions	7
2 Derivation of Lane-Emden from Variation of Energy Principle	8
2.1 Energy of the System	8
2.2 Interpretation	9
2.3 Euler-Lagrange with Lagrange Multiplier	10
3 Numerical and Analytical Analysis	12
3.1 Analytical Solutions	12
3.1.1 Case 1: Polytropic Index $n = 0$	12
3.1.2 Case 2: Polytropic Index $n = 1$	13
3.1.3 Case 3: Polytropic Index $n = 5$	14
3.1.4 Special Case: Polytropic Index $n = 2$	15
3.2 Numerical Analysis	17
3.3 Comparison of Lane-Emden Model with Stellar Observational Data	20
3.3.1 Radius Comparison	20
3.3.2 Mass Comparison	21
3.3.3 Density Profile	22
4 Conclusion	23
Bibliography	23

Chapter 1. Derivation of the Lane-Emden from First Principles

In order to introduce the nature of the Lane-Emden Equation (LEE), it is useful to first understand the derivation and the assumptions taken during the process.

§1.1 Assumptions

To begin, we can assume that there is a non-rotating stationary/static spherical body of radius R mass M . It has a variable density which is characterized by a function $\rho = \rho(X_i)$, specified by some arbitrary variables X_i . Similarly, the mass it contains at a certain point by some arbitrary variables is defined as $M = M(X_i)$. X_i denotes the variables which the functional depends on. Additionally, the matter of the spherical body is considered to be a continuous, non-compressible fluid such that the functionals we have defined are continuous and differentiable.

These are some facts which arise from combining the above assumptions.

1. Density ρ is a function of radius r only: $\rho = \rho(r)$
2. Mass M is a function of radius r only: $M = M(r)$
3. Only the inner mass contributes to the gravitational force.

The above are implications which arise from Shell Theorem.

On a general note, the above assumptions removes complexities which may lead to more complex but more accurate models.

In order to keep simplicity, Newtonian gravitation is employed.

$$F_{12} = \frac{GM_1M_2}{r_{12}^2} \quad (1)$$

The more useful form for our derivation is the differential form:

$$dF_{12} = \frac{GM_1}{r_{12}^2} dM_2 \quad (2)$$

Very importantly, we are ignoring any relativistic effects that the Stellar body may have.

Additionally we will consider a thin shell during the computation. This will allow us to make the following assumptions:

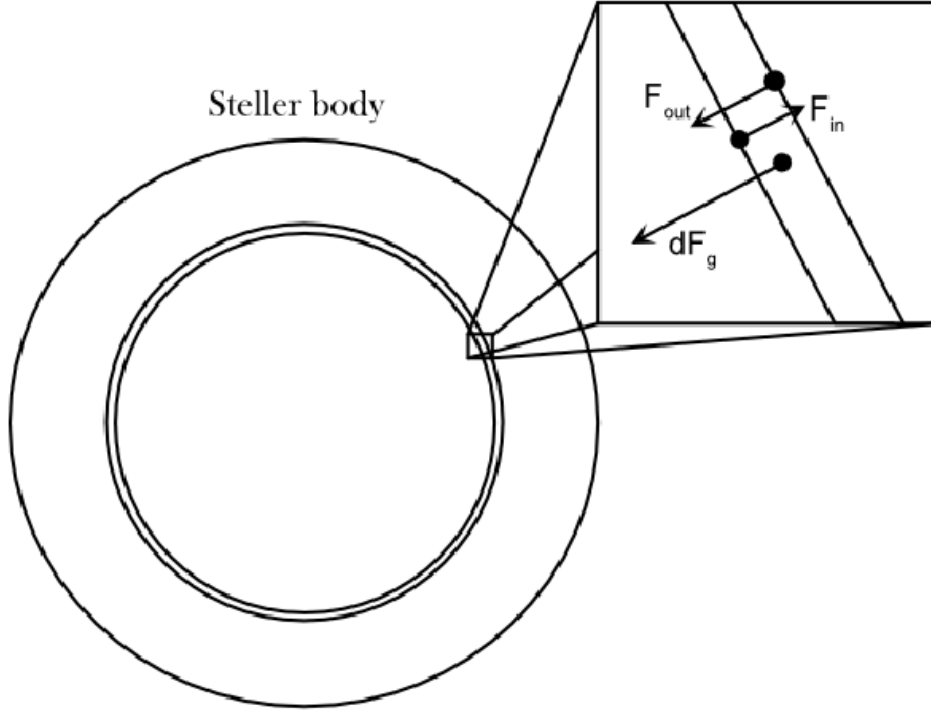
1. Higher orders of differentials (like $(dr)^2$) can be ignored.
2. Change of pressure from $P(r)$ to $P(r + dr)$ can be approximated as $P(r)$ to $P(r) + dP$.
3. Density is constant across the thin shell.

Additionally, Stellar body processes like nuclear reactions, which certainly do contribute to the factors pertaining to Stellar bodies. Later on, we will discuss that trying to include this factor does lead to more accurate results. In the next section, we discuss the derivation of the hydrostatic equilibrium equation, which can be considered to be the very base of the LEE.

§1.2 Hydrostatic Equilibrium

We can deduce that at some arbitrary thin shell of a spherical body, the pressure and gravity due to the inner mass must be in equilibrium at every point, as the stellar body itself is in equilibrium. The mathematical statement of this is the hydrostatic equilibrium equation.

Assume that at a radius of r from the center, there is a thin shell of radius dr . The thin shell has some pressure on it in its inner surface $P_{\text{in}} = P(r)$ and some pressure on its outer surface $P_{\text{out}} = P(r) + dP$.



The area on which these pressures act on are $A_{\text{in}} = 4\pi r^2$ and $A_{\text{out}} = 4\pi(r + dr)^2$ respectively. Thus we find:

$$F_{\text{in}} = P_{\text{in}} A_{\text{in}} = P(r)(4\pi r^2) \quad (3)$$

$$F_{\text{out}} = P_{\text{out}} A_{\text{out}} = (P(r) + dP)[4\pi(r + dr)^2] = 4\pi[r^2 P(r) + 2rdrP(r) + (dr)^2 P(r) + r^2 dP(r) + 2rdrdP(r) + (dr)^2 dP(r)] \quad (4)$$

We can ignore the higher order of differentials which includes 2nd and 3rd order terms like $drdP(r)$, $(dr)^2$ and $(dr)^2 dP$ which do not contribute to the force on the same order as the 1st order.

$$F_{\text{out}} \approx 4\pi r^2 P(r) + 4\pi r^2 dP(r) \quad (5)$$

In order to find the force due to gravity, we can use the differential form of Newton's law of gravitation:

$$dF_g = \frac{GM(r)dM}{r^2} \quad (6)$$

Here F_g is the force due to gravity, G is the gravitational constant where $G = 6.6743 \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, dm is the mass of the thin shell and r is the radius from the center of the stellar body.

We know that:

$$\rho(r) = \frac{dM}{dV} \quad (7)$$

where ρ is the density which is a function of radius r only, dm is the mass of the thin shell, dV is the thin volume of the shell. It is quite evident that:

$$dV = d\left(\frac{4}{3}\pi r^3\right) = 4\pi r^2 dr \quad (8)$$

Using Equation 8 in Equation 7, we can obtain:

$$dM = 4\pi r^2 \rho(r) dr \quad (9)$$

Using Equation 9 in Equation 6 we further get:

$$dF_g = 4\pi GM(r)\rho(r)dr \quad (10)$$

We invoke the condition that the stellar body is in equilibrium, thus the sum of all forces must be equal to 0.

$$\sum_i \vec{F}_i = 0 \quad (11)$$

Considering that any vector going radially inwards is assigned a negative sign. We only have the following forces assuming that there is no rotational forces as well, considering that the body is not spinning.

$$F_{\text{out}} - F_{\text{in}} - dF_g = 0 \quad (12)$$

We can write each term using Equation 3, Equation 5, Equation 10:

$$4\pi r^2 P(r) + 4\pi r^2 dP(r) - (4\pi r^2 P(r)) - (4\pi GM(r)\rho(r)dr) = 0 \quad (13)$$

Rearranging the terms we obtain:

$$-\frac{dP(r)}{dr} = \frac{GM(r)\rho(r)}{r^2} \quad (14)$$

Additionally using Equation 9, we can rearrange to obtain a differential equation which describes how the mass changes with respect to radius.

$$\frac{dM}{dr} = 4\pi r^2 \rho(r) \quad (15)$$

§1.3 Polytropic Equation of State

$$PV^\gamma = \text{constant} \quad (16)$$

This is a result that is derived from thermodynamics, assuming that there is an adiabatic condition (). We will apply this condition to our stellar body. The interesting part is how even though the equation is simple, it does tend to resemble real-life stellar masses for specific values of γ . It has been found that for $\gamma = \frac{4}{3}$ it actually corresponds to something called the “Eddington Standard Model”, which is closely related to describing a star in radiative equilibrium [1]. Perhaps more interestingly, for $\gamma = \frac{5}{3}$, it can describe (although crudely) a white dwarf. It may be evident that these values of γ are a little strange to work with so we will introduce a number called polytropic index n . Such that $\gamma = 1 + \frac{1}{n}$.

§1.4 Lane Emden Equation

Now that we have found the necessary equations needed to describe our system cleanly, we can derive something called the Lane-Emden Equation [2], [3]. This is quite a famous equation as it is one of the rare cases where an analytical solution can be found for a stellar body. Starting with Equation 14:

$$-\frac{dP(r)}{dr} = \frac{GM(r)\rho(r)}{r^2} \quad (17)$$

We can rearrange it as follows:

$$\frac{r^2}{\rho(r)} \frac{dP(r)}{dr} = -GM(r) \quad (18)$$

Differentiating with respect to r :

$$\frac{d}{dr} \left(\frac{r^2}{\rho(r)} \frac{dP(r)}{dr} \right) = -G \frac{dM(r)}{dr} \quad (19)$$

We can use Equation 15 in Equation 19 and rearrange:

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho(r)} \frac{dP(r)}{dr} \right) = -4\pi G \rho(r) \quad (20)$$

Now for the interesting part, the above equation can be simplified in an extremely smart manner, we will introduce dimensionless variables θ and ξ :

$$\rho(r) \equiv \rho_c \theta^n \quad (21)$$

$$r \equiv \alpha \xi \quad (22)$$

Here θ varies between 0 and 1: $0 \leq \theta \leq 1$. Additionally, ρ_c is initially just a constant variable, however, it is found that this is actually the central density of the star, α is a variable which will be chosen in order to simplify the equation further.

We can differentiate two above equations implicitly to obtain:

$$d\rho = \rho_c n \theta^{n-1} d\theta, \quad dr = \alpha d\xi \quad (23)$$

Dividing both we can obtain:

$$\frac{d\rho}{dr} = \frac{\rho_c n \theta^{n-1}}{\alpha} \frac{d\theta}{d\xi} \quad (24)$$

In Equation 20, we can use the polytropic EOS Equation 16 to obtain:

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho(r)} \frac{d(K\rho^\gamma)}{dr} \right) = -4\pi G \rho(r) \quad (25)$$

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{K r^2 \gamma \rho^{\gamma-1}}{\rho(r)} \frac{d\rho}{dr} \right) = -4\pi G \rho(r) \quad (26)$$

We can now use Equation 24, Equation 21 and Equation 22 to write:

$$\frac{1}{\alpha^2 \xi^2} \frac{d}{d\xi} \left[\frac{K \alpha^2 \xi^2 \gamma \rho_c^{\gamma-1} \theta^{n(\gamma-1)}}{\rho_c \theta^n} \left(\frac{\rho_c n \theta^{n-1}}{\alpha^2} \frac{d\theta}{d\xi} \right) \right] = -4\pi G \rho_c \theta^n \quad (27)$$

We can simplify the equation also using the relation $\gamma = 1 + \frac{1}{n}$ in order to obtain:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\alpha^2 \left[\frac{4\pi G \rho_c}{K(n+1)\rho_c^{\frac{1}{n}}} \right] \theta^n \quad (28)$$

We chose α as:

$$\alpha \equiv \left[\frac{K(n+1)\rho_c^{\frac{1}{n}}}{4\pi G \rho_c} \right]^{\frac{1}{2}} \quad (29)$$

Thus we can obtain the final equation:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n \quad (30)$$

This is called the Lane-Emden Equation.

§1.5 Boundary Conditions

In order to solve this 2nd order differential equation, we will require some boundary conditions such that they make physical sense. The most commonly used boundary conditions use the following ideas:

1. The density of the matter of the star at its outer surface must be 0.
2. At the center of the star, the pressure gradient goes to 0 (imagining the pressure gradient as a vector, at the center there is no vector as there is an equal pressure gradient from all directions due to the symmetry of the system.)
3. At the center of the star, there is a central density ρ_c .

Each of these statements are valid on our physical system, thus are valid boundary conditions. We can now rewrite these as mathematical statements respectively

1. At the outer surface of the star ξ_1 , $\theta(\xi_1) = 0$.
2. At center of the star $\xi = 0$, $\frac{d\theta}{d\xi} = 0$.
3. At the center of the star $\rho = \rho_c$, from Equation 21 we obtain: $\theta(0) = 1$.

As this is a 2nd order ODE, two boundary conditions are required to determine a unique solution. Condition 2 and 3 are the natural choices as they both specify the condition at the center of the star, whereas condition 1 emerges from the solution itself.

Chapter 2. Derivation of Lane-Emden from Variation of Energy Principle

In the section before, we had used Newtonian mechanics in order to obtain an equation which describes the stellar body. This is a direct and physically intuitive derivation of the Lane-Emden equation. However, in this section, we will try to obtain the Lane-Emden using a more Lagrangian framework [4]. In the Newtonian framework we justify stability of stellar bodies using force analysis. In the Lagrangian framework, we will justify the stability of the stellar body by assuming that the star would be at the minima of its total energy of the stellar system. Both analysis are correct, however the Lagrangian framework guarantees that the stellar body has stability in terms of its energy as any small perturbations in energy due to any displacements will still guarantee stability and the system returns to equilibrium. In a force balancing model, there is no such guarantee, it could be a saddle point or even a maximum. Additionally, this confirms that the Lane-Emden is a fundamentally physical description.

§2.1 Energy of the System

The total energy of the system comes from two sources: the gravitational energy and the thermal energy contained in the system. We write:

$$E_{\text{total}} = E_{\text{gravitation}} + E_{\text{thermal}} \quad (31)$$

In order to determine the total energy, we need to write appropriate relations for $E_{\text{gravitation}}$ and E_{thermal} for our system. We know the contribution of change in gravitational potential for Newtonian gravitation for each spherical shell to be:

$$dU_{\text{gravitation}} = -\frac{GM}{r}dM \quad (32)$$

Considering each spherical shell we obtain the energy contributed by gravity:

$$E_{\text{gravitation}} = -\int_0^R \frac{GM}{r}dM \quad (33)$$

R denotes the radius of the spherical stellar body. Writing in a more useful form and using Equation 15:

$$E_{\text{gravitation}} = -\int \frac{GM}{r} \frac{dM}{dr} dr = -\int 4\pi G \rho(r) M(r) r dr \quad (34)$$

We also know that the absolute internal energy of an ideal gas to be:

$$U = \frac{f}{2} PV \quad (35)$$

Where f denotes a quantity known as degrees of freedom. We write the following differential form considering the thermal energy of each spherical shell:

$$dU_{\text{thermal}} = \frac{f}{2} P dV = \frac{P}{\gamma - 1} dV = \frac{P}{\gamma - 1} 4\pi r^2 dr = \frac{4\pi K}{\gamma - 1} r^2 \rho^\gamma dr \quad (36)$$

We consider each spherical shell and obtain the following thermal energy:

$$E_{\text{thermal}} = \int_0^R \frac{4\pi K}{\gamma - 1} r^2 \rho^\gamma dr \quad (37)$$

We write the final form of the energy as:

$$E_{\text{total}} = \int_0^R \left(\frac{4\pi K}{\gamma - 1} r^2 \rho^\gamma - 4\pi G \rho(r) M(r) r \right) dr \quad (38)$$

We will consider that minimising the energy and thus treating it as the analog of action, will end up obtaining an EOS for the stellar body. This is the principle of the following derivation.

§2.2 Interpretation

We require to use Lagrangian Field Theory in order to study the behaviour of this system properly. Here instead of using the standard $\mathcal{L}$ Lagrangian it is replaced by a Lagrangian density. The explicit difference is in the way that both are defined. The lagrangian operates over a finite amount of generalised coordinates which is mathematically described as:

$$\mathcal{L} = L(q, \dot{q}, t) \quad (39)$$

Where the action S is:

$$S[q] = \int_{t_0}^{t_1} L(q, \dot{q}, t) dt \quad (40)$$

However, we are instead working with functionals which resemble scalar fields. In such cases it is much more common to work with the Lagrangian density which is represented as $\mathcal{L}$ as well, since it is commonly referred to as the Lagrangian as well, in this specific definition:

$$\mathcal{L}\left(\varphi_i, \nabla \varphi_i, \frac{\partial \varphi_i}{\partial t}, \mathbf{X}, t\right) \sim \mathcal{L}(\varphi_i, \nabla \varphi_i, \mathbf{X}) \quad (41)$$

The removal of the partial differentiation of φ_i reflects that there is no time-evolution in the system. The system we are studying is inherently static. Where the action functional $\mathcal{S}$ now is defined as:

$$\mathcal{S}[\varphi_i] = \int \mathcal{L}(\varphi_i, \nabla \varphi_i, \mathbf{X}, t) d^3x dt \quad (42)$$

Consider that the action functional is the analog of the total energy that we had described. Actually, it is more valid to refer our “action” as the total energy functional instead of action function. The validity of the Euler-Lagrange is not only valid for action functionals, letting us find extrema for our energy functional finds the minimum energy system of our static stellar body system. We will also consider that the time analog is the radius. Interesting, both time in the action functional for a standard lagrangian density and the radius we have described up to now are both independent variables, which describe the “path” of the action. Thus, they are fitting analogs. We can now write:

$$\mathcal{S}[\varphi_i] = \int \mathcal{L}\left(\varphi_i, \nabla \varphi_i, \frac{\partial \varphi_i}{\partial t}, \mathbf{X}, t\right) d^3x dt \sim \mathcal{E}[\rho, M] = \int \mathcal{K}(\rho, \nabla \rho, M, \nabla M, r) dr \quad (43)$$

Where $\mathcal{K}(\rho, \nabla \rho, M, \nabla M, r) = \mathcal{L}(\rho, \nabla \rho, M, \nabla M, r) dV$. It is important to note that as our system is spherically symmetric and time-independent. The spatial integral will reduce from $d^3x \rightarrow 4\pi \int r^2 dr$. Additionally, as this is time-independent, there is no dependence of the energy functional on time t . Thus, our functional reduces to the form of $\int \mathcal{K} dr$.

§2.3 Euler-Lagrange with Lagrange Multiplier

We now apply the Euler-Lagrange equation with a caveat, M is functionally related to ρ by the constraint $dM/dr = 4\pi r^2 \rho(r)$. In order to properly apply this to our case, we can assume that M and ρ are not functionally related and introduce a Lagrange multiplier $\lambda(r)$ and write the following:

$$\mathcal{F} = \left[\frac{4\pi K \rho^\gamma r^2}{\gamma - 1} - 4\pi G \rho(r) M(r) r + \lambda(r) \left[\frac{dM}{dr} - 4\pi r^2 \rho \right] \right] \quad (44)$$

Differentiating Equation 44 with respect to ρ and using the following Euler-Lagrange of $\rho(r)$ (The second term is 0 as $\mathcal{F}$ as has no dependence on ρ'):

$$\frac{\partial \mathcal{F}}{\partial \rho} - \frac{d}{dr} \left(\frac{\partial \mathcal{F}}{\partial \rho'} \right) = 0 \quad (45)$$

$$\frac{\partial \mathcal{F}}{\partial \rho} = \left[\frac{4\pi K \gamma \rho^{\gamma-1} r^2}{\gamma - 1} - 4\pi G M(r) r - 4\pi \lambda(r) r^2 \right] = 0 \quad (46)$$

Differentiating Equation 44 with respect to M :

$$\frac{\partial \mathcal{F}}{\partial M} = -4\pi G \rho(r) r \quad (47)$$

Then differentiating Equation 44 with respect to M' and further differentiating with respect to r and then applying the Euler-Lagrange for the mass:

$$\frac{\partial \mathcal{F}}{\partial M'} = \lambda(r) \quad (48)$$

$$\frac{d}{dr} \left(\frac{\partial \mathcal{F}}{\partial M'} \right) = \lambda'(r) \quad (49)$$

$$\frac{\partial \mathcal{F}}{\partial M} - \frac{d}{dr} \left(\frac{\partial \mathcal{F}}{\partial M'} \right) = 0 \quad (50)$$

We obtain the following:

$$\lambda'(r) = -4\pi G \rho(r) r \quad (51)$$

We can rewrite Equation 46 such that λ is isolated:

$$\lambda(r) = \frac{K\gamma}{\gamma - 1} \rho^{\gamma-1} - \frac{GM}{r} \quad (52)$$

Differentiating with respect to r :

$$\lambda' = K\gamma \rho^{\gamma-2} \rho' - G \left(\frac{dM}{dr} \frac{1}{r} - \frac{M}{r^2} \right) \quad (53)$$

We can use Equation 51 in Equation 53, additionally using Equation 15 we can obtain the following by isolating M

$$M = -\frac{K\gamma}{G} \rho^{\gamma-2} \rho' r^2 \quad (54)$$

Differentiating Equation 54 with respect to r :

$$M' = -\frac{K\gamma}{G} \frac{d}{dr} (\rho^{\gamma-2} \rho' r^2) \quad (55)$$

Rearranging and using Equation 15:

$$\frac{1}{r^2} \frac{d}{dr} (r^2 \rho' \rho^{\gamma-2}) = -\frac{4\pi G}{K\gamma} \rho \quad (56)$$

Now using the non-dimensionalisation step from Equation 21, Equation 22 and Equation 24, additionally using $\gamma = 1 + \frac{1}{n}$:

$$\frac{1}{(\alpha\xi)^2} \cdot \frac{1}{\alpha} \frac{d}{d\xi} \left((\alpha\xi)^2 \left(\frac{\rho_c n \theta^{n-1}}{\alpha} \frac{d\theta}{d\xi} \right) (\rho_c^{\gamma-2} \theta^{1-n}) \right) = -\frac{4\pi G}{K\gamma} \rho_c \theta^n \quad (57)$$

Simplifying and rearranging we obtain:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\alpha^2 \left[\frac{4\pi G \rho_c}{K(n+1) \rho_c^{\frac{1}{n}}} \right] \theta^n \quad (58)$$

We can define α as:

$$\alpha \equiv \left[\frac{K(n+1) \rho_c^{\frac{1}{n}}}{4\pi G \rho_c} \right]^{\frac{1}{2}} \quad (59)$$

We obtain the Lane-Emden again:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n \quad (60)$$

Chapter 3. Numerical and Analytical Analysis

After deriving the Lane-Emden equation, we now look for solutions with physically meaningful values of the polytropic index n . Exact analytical solutions exist only for $n = 0, 1$, and 5 . Interestingly, there also exists a series solution for $n = 2$. For other values, including the physically important cases $n = 1.5$ and $n = 3$, which model fully convective and radiative stars respectively. Do not have closed-form solutions and numerical methods are required. We restrict our analysis to $0 \leq n \leq 5$, since $n \geq 5$ produces stars of infinite radius with limited physical relevance.

§3.1 Analytical Solutions

These are the cases where the Lane-Emden can be solved by using analytical techniques.

§3.1.1 Case 1: Polytropic Index $n = 0$

When $n = 0$, the term $\theta^n \rightarrow 1$, thus we write:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -1 \quad (61)$$

Rearranging:

$$\frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\xi^2 \quad (62)$$

$$\xi^2 \frac{d\theta}{d\xi} = - \int \xi^2 d\xi \quad (63)$$

$$\xi^2 \frac{d\theta}{d\xi} = -\frac{1}{3}\xi^3 + C_1 \quad (64)$$

$$\frac{d\theta}{d\xi} = -\frac{1}{3}\xi + \frac{C_1}{\xi^2} \quad (65)$$

Integrating again:

$$\theta(\xi) = -\frac{1}{6}\xi^2 - \frac{C_1}{\xi} + C_2 \quad (66)$$

We can use our initial conditions which we had discussed before, namely conditions 2 and 3:

2. At center of the star $\xi = 0$, $\frac{d\theta}{d\xi} = 0$ or $\theta' = 0$.
3. At the center of the star $\rho = \rho_c$, from Equation 21 we obtain: $\theta(0) = 1$.

From these we obtain:

$$\theta(\xi) = 1 - \frac{1}{6}\xi^2 \quad (67)$$

§3.1.2 Case 2: Polytropic Index $n = 1$

When $n = 1$, the term $\theta^n \rightarrow \theta$, thus we can write:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta \implies \theta'' + \frac{2}{\xi} \theta' + \theta = 0 \quad (68)$$

Assuming that a power series solution exists:

$$\theta(\xi) = \sum_{n=0}^{\infty} a_n \xi^n \quad (69)$$

Finding θ' and θ'' :

$$\theta'(\xi) = \sum_{n=1}^{\infty} n a_n \xi^{n-1}, \quad \theta''(\xi) = \sum_{n=2}^{\infty} n(n-1) a_n \xi^{n-2} \quad (70)$$

Substituting into the expanded form of the Lane-Emden:

$$\sum_{n=2}^{\infty} n(n-1) a_n \xi^{n-2} + 2 \sum_{n=1}^{\infty} n a_n \xi^{n-2} + \sum_{n=0}^{\infty} a_n \xi^n = 0 \quad (71)$$

We can simplify it to:

$$\sum_{n=2}^{\infty} n(n+1) a_n \xi^{n-2} + \sum_{n=0}^{\infty} a_n \xi^n = 0 \quad (72)$$

We can shift the first term such that it is in the form of $\sum_{k=0}^{\infty}$, assume that $k = n - 2$:

$$\sum_{n=2}^{\infty} n(n+1) a_n \xi^{n-2} = \sum_{k=0}^{\infty} (k+2)(k+3) a_{k+2} \xi^k \quad (73)$$

Combining into a single series:

$$\sum_{k=0}^{\infty} [(k+2)(k+3) a_{k+2} + a_k] \xi^k = 0 \quad (74)$$

For power series to be equal to 0, all the coefficients must also be 0:

$$(k+2)(k+3) a_{k+2} + a_k = 0 \quad (75)$$

Rearranging:

$$a_{k+2} = -\frac{a_k}{(k+2)(k+3)} \quad (76)$$

We can apply the boundary conditions which we have discussed before to find that:

$$a_0 = 1, a_1 = 0 \quad (77)$$

However, we find $a_2, a_3, a_4 \dots$ by the recursive relation we found. We can write out the full equation and we find that:

$$\theta(\xi) = 1 - \frac{\xi^2}{6} + \frac{\xi^4}{120} - \frac{\xi^6}{5040} + \dots \quad (78)$$

We will find that:

$$\sin(\xi) = \xi - \frac{\xi^3}{6} + \frac{\xi^5}{120} - \frac{\xi^7}{5040} + \dots \Rightarrow \frac{\sin(\xi)}{\xi} = 1 - \frac{\xi^2}{6} + \frac{\xi^4}{120} - \frac{\xi^6}{5040} + \dots \quad (79)$$

Thus:

$$\theta(\xi) = \frac{\sin(\xi)}{\xi} \quad (80)$$

§3.1.3 Case 3: Polytropic Index $n = 5$

For $n = 5$, the term $\theta^n \rightarrow \theta^5$, thus we can write:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^5 \Rightarrow \theta'' + \frac{2}{\xi} \theta' + \theta^5 = 0 \quad (81)$$

Assume that the functional form of $\theta(\xi)$ is quadratic:

$$\theta(\xi) = (1 + a\xi^2)^b \quad (82)$$

We can find θ' and θ'' :

$$\theta' = 2ab\xi(1 + a\xi^2)^{b-1}, \quad \theta'' = 2ab(1 + a\xi^2)^{b-1} + 4a^2b(b-1)\xi(1 + a\xi^2)^{b-2} \quad (83)$$

We substitute it into the expanded Lane-Emden, and after some simplification:

$$(1 + a\xi^2)^{b-2} [6ab(1 + a\xi^2) + 4a^2b(b-1)\xi^2] + (1 + a\xi^2)^{5b} = 0 \quad (84)$$

For this to be true, all the co-efficients of $(1 + a\xi^2)$ should match, which obtains:

$$b - 2 = 5b \Rightarrow b = -\frac{1}{2} \quad (85)$$

Which gives:

$$\theta(\xi) = \frac{1}{\sqrt{1 + a\xi^2}} \quad (86)$$

We can now substitute $b = -\frac{1}{2}$ into Equation 84 and simplify to obtain:

$$(1 + a\xi^2)^{-\frac{5}{2}} - 3a(1 + a\xi^2)^{-\frac{5}{2}} = 0 \quad (87)$$

Factoring:

$$(1 - 3a)(1 + a\xi^2)^{-\frac{5}{2}} = 0 \quad (88)$$

As the quadratic cannot be 0 or negative:

$$a = \frac{1}{3} \quad (89)$$

Thus we obtain:

$$\theta(\xi) = \left(1 + \frac{\xi^2}{3} \right)^{-\frac{1}{2}} \quad (90)$$

§3.1.4 Special Case: Polytopic Index $n = 2$

As $n = 2$, the term $\theta^n \rightarrow \theta^2$, we can write:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^2 \implies \theta'' + \frac{2}{\xi} \theta' + \theta^2 = 0 \quad (91)$$

Assuming that a power series solution exists around $\xi = 0$:

$$\theta(\xi) = \sum_{n=0}^{\infty} a_n \xi^n \quad (92)$$

Finding θ' and θ'' :

$$\theta'(\xi) = \sum_{n=1}^{\infty} n a_n \xi^{n-1}, \quad \theta''(\xi) = \sum_{n=2}^{\infty} n(n-1) a_n \xi^{n-2} \quad (93)$$

Substituting into the expanded Lane-Emden:

$$\sum_{n=2}^{\infty} n(n+1) a_n \xi^{n-2} + \left(\sum_{n=0}^{\infty} a_n \xi^n \right)^2 = 0 \quad (94)$$

We can change the index of the left term and apply the Cauchy product on the right term:

$$\sum_{k=0}^{\infty} (k+2)(k+3) a_{k+2} \xi^k + \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j a_{k-j} \right) \xi^k = 0 \quad (95)$$

We can combine the series to obtain:

$$\sum_{k=0}^{\infty} \left[(k+2)(k+3) a_{k+2} + \sum_{j=0}^k a_j a_{k-j} \right] \xi^k = 0 \quad (96)$$

As the series converges to 0, it must also only have coefficients of value 0:

$$(k+2)(k+3) a_{k+2} + \sum_{j=0}^k a_j a_{k-j} = 0 \quad (97)$$

Rearranging we obtain the following recurrence:

$$a_{k+2} = -\frac{1}{(k+2)(k+3)} \sum_{j=0}^k a_j a_{k-j} \quad (98)$$

From the boundary conditions putting it in the Lane-Emden:

$$\theta(0) = 1, \theta'(0) = 0 \rightarrow a_0 = 1, a_1 = 0 \quad (99)$$

We can now calculate different values of index $j = 1, 3, 5 \dots$ focusing on odd terms:

$$a_3 = -\frac{1}{12} (a_0 a_1 + a_1 a_0) = 0 \quad (100)$$

$$a_5 = -\frac{1}{30} (a_0 a_3 + a_1 a_2 + a_2 a_1 + a_3 a_0) = 0 \quad (101)$$

Turns out, by method of induction, it can be shown that $a_{2k+1} = 0$

Introducing a series such that: $b_n := a_{2n}$, the power series becomes:

$$\theta = \sum_{n=0}^{\infty} b_n \xi^{2n} \quad (102)$$

In the recurrence relation, we can set $k = 2n$, the recurrence becomes:

$$a_{2n+2} = -\frac{1}{(2n+2)(2n+3)} \sum_{j=0}^{2n} a_j a_{2n-j} \quad (103)$$

Notice that the odd coefficients are 0 and thus can set $k = 2r$ and use $b_r = a_{2r}$, we obtain:

$$b_{n+1} = -\frac{1}{(2n+2)(2n+3)} \sum_{r=0}^n b_r b_{n-r} \quad (104)$$

By using the initial conditions and the recursive relation we can obtain a series:

$$\theta(\xi) = 1 - \frac{\xi^2}{6} + \frac{\xi^4}{60} - \frac{11\xi^6}{7560} + \dots \quad (105)$$

It is important to establish that this is not an analytical solution and will only give better approximations when the series is expanded for higher order. Interestingly, the radius of the physical stellar body is at $\xi_1 \approx 4.3528$. Moreover, this series converges for $\xi \leq 15.7179$, this is due to the fact that there exists two singularities at $\xi \approx \pm 15.7179i$, thus the radius of convergence is not infinite. Deriving this is not taken up as additional numerical analysis is required for it.

§3.2 Numerical Analysis

I will be using Python 3.13.11, numpy, matplotlib and scipy in order to carry out the numerical analysis. Using the `solve_ivp` module from scipy in order to solve the ODE. The reason for this choice is nothing other than just familiarity for me. I assume any other method such as `odeint` should also work.

As the Lane-Emden is a 2nd order ODE, we will need to represent it instead as a coupled 1st order ODE instead as `solve_ivp` is unable to solve 2nd order ODEs. In order to do so consider the two substitutions:

$$u_1 = \theta, u_2 = \frac{d\theta}{d\xi} \quad (106)$$

The Lane-Emden can thus be decomposed to:

$$\frac{du_1}{d\xi} = u_2, \quad \frac{du_2}{d\xi} = -u_1^n - \frac{2}{\xi}u_2 \quad (107)$$

The code which encodes the above is taken up by a defined function called the *lane_emden*:

```
def lane_emden(xi, J, n):
    u1, u2 = J
    if u1 <= 0:
        return [u2, 0]
    return [u2, -(u1**n) - 2*u2/xi]
```

Another thing to note is that the term u_1^n will have to be dealt with more caution as when working in Python, if u_1 becomes a negative quantity, we can face issues with the numerical process as if n is not an integer value, then the calculation can break. The way I dealt with this is by doing the following: adding the *if* statement in the above code, which returns the vector J as $[u_2, 0]$; introducing a *surface.at* event which stops the process of calculation at the first instance of $\theta = 0$.

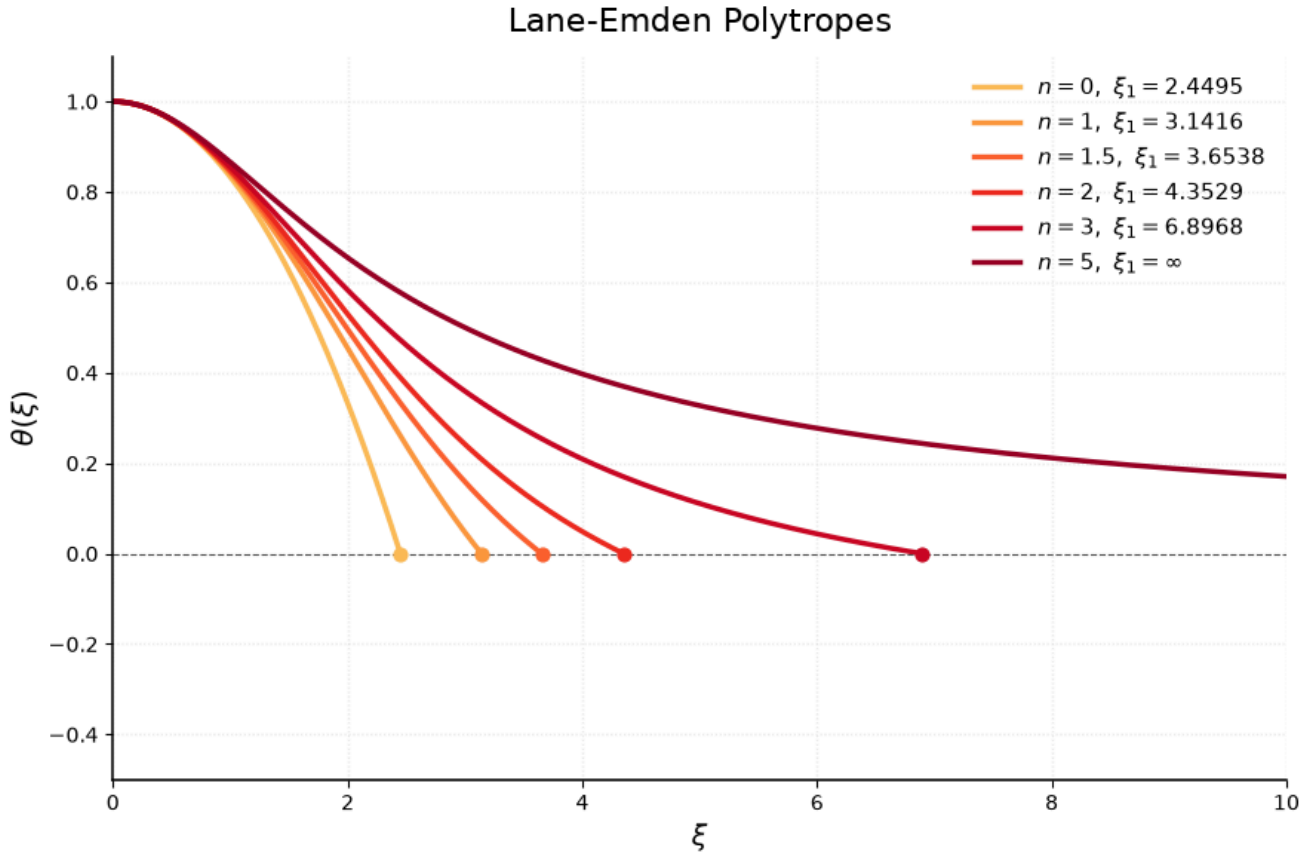
```
def surface_at(xi, J, n):
    u1, u2 = J
    return u1
```

```
surface_at.terminal = True
surface_at.direction = -1
```

The way this works is when the value of u_1 becomes 0, it returns *False* which makes triggers the *surface.at* event and it has to be manually set that the *.terminal* property of the event is to be set to *True*. The *.direction* property is used to detect how exactly the crossing event occurs, in this case, it must be always be such that $\theta > 0 \rightarrow \theta < 0$.

```
def solve(n):
    xi_i = 1e-8
    xi_f = 80
    u1_0 = 1 - xi_i**2 / 6
    u2_0 = -xi_i / 3
    sol = solve_ivp(
        lane_emden,
        [xi_i, xi_f],
        [u1_0, u2_0],
        events = surface_at,
        args = (n,),
        dense_output = True,
        max_step = 0.01
    )
    return sol
```

The above `solve` function take a value of n and uses the `lane_emden` function as one of its parameters and solves it using the `solve_ivp` module from `scipy`. Internally, it uses the Runge-Kutta method of order 5(4). This is an iterative numerical technique which is used to solve ODEs. Exact information about the function is documented in the python file itself. The rest of the file deals with just the appearance and how the graph is represented. Using the code, we obtain the following graph:



Interestingly, the case of the $n = 5$ has an infinite radius, this actually shows that the $n = 5$ polytrope and beyond do not have any physical relevance. However, certain special cases do. Such as $n = 1.5$, which model the non-relativistic degenerate electron masses such as fully convective low-mass stars and low-mass white dwarves. For $n = 3$, it can model relativistic white-dwarves or radiation pressure dominated star.

However, as we will see shortly, these polytropes are only approximations and have their limitations.

```
PS C:\Users\arush\hello\Manimtime> python -u "c:\Users\arush\hello\Manimtime\lane_emden solver.py"
0 xi_1=2.449489864129675 [dtheta/dxi at xi_1 = -0.8161581907419493] [-xi_1^2 dtheta/dxi at xi_1 = 4.8969496296365325 ]
1 xi_1=3.1415935230413856 [dtheta/dxi at xi_1 = -0.31785440168527485] [-xi_1^2 dtheta/dxi at xi_1 = 3.1370989381937604 ]
1.5 xi_1=3.6537536953613796 [dtheta/dxi at xi_1 = -0.20331801407288616] [-xi_1^2 dtheta/dxi at xi_1 = 2.7142784226534418 ]
2 xi_1=4.352874665676004 [dtheta/dxi at xi_1 = -0.12731824306313463] [-xi_1^2 dtheta/dxi at xi_1 = 2.412364683716666 ]
3 xi_1=6.896848828337062 [dtheta/dxi at xi_1 = -0.04242655711309242] [-xi_1^2 dtheta/dxi at xi_1 = 2.0180838370145473 ]
5 xi_1=[]
```

n	ξ_1	$\theta'(\xi_1)$	$-\xi_1^2\theta'(\xi_1)$
0	2.44949	-0.81658	4.89695
1	3.14159	-0.31785	3.13710
1.5	3.65375	-0.20332	2.71428
2	4.35287	-0.12732	2.41236
3	6.89685	-0.04243	2.01808

Table 1: Numerically computed Lane-Emden constants.

§3.3 Comparison of Lane-Emden Model with Stellar Observational Data

§3.3.1 Radius Comparison

From the numerical analysis, we conducted in the previous section. We have found the radius of the surface and the distribution of the stellar mass. In order to compare what the ability of the Lane-Emden equation is able to predict with respect to stellar observations, we need to make certain assumptions and then see if it stays valid for other quantities.

To do so, we must convert the ξ back to radius r . Once, this has been calculated we can actually start to compare it to real stellar observation. We know that $r = \alpha\xi$. Correctly, determining the value of α will determine the value of r . Previously from the relationship of α we found before:

$$\alpha \equiv \left[\frac{K(n+1)\rho_c^{\frac{1}{n}}}{4\pi G\rho_c} \right]^{\frac{1}{2}} \quad (108)$$

In order to get started on our comparative work, let us take the polytropic index $n = 3$, which represents a radiation pressure dominated star/a relativistic white dwarf:

$$\alpha = \left[\frac{K}{\pi G\rho_c^{2/3}} \right]^{\frac{1}{2}} \quad (109)$$

Here we have to determine both ρ_c and K , in order to do so we can use one of the values from a standard model called the “Model S” [5], which is sourced from the Christensen-Dalsgaard solar model data and we try to see if another values follow the same standard value. The difference of those values from the standard and the extent to which it is does, will give an idea for how accurate the Lane-Emden equation is.

Assuming the following value from the Model S, that the $\rho_c = 1.622 \times 10^5 \text{ kg/m}^3$ and that $P_c = 2.477 \times 10^{16} \text{ Pa}$. Using this we can determine α and K as $K = P_c/\rho_c^{4/3}$ at the centre.

Computing we find, $\alpha = 6.707 \times 10^7 \text{ m}$. We know that the surface of a $n = 3$ star is determined to be $\xi_1 = 6.897$ from the numerical analysis. We can determine the radius of the surface from $r = \alpha\xi$ to be $r = 4.625 \times 10^8 \text{ m}$. This is the radius that is determined by the Lane-Emden Model. However, the standard value of $R_\odot$ is determined to be $R_\odot = 6.957 \times 10^8 \text{ m}$. The error which the Lane-Emden has as compared to observational data is calculated from the following:

$$\begin{aligned} \text{\%Error in Radius Determination} &= \frac{|R_\odot - r|}{R_\odot} \times 100 \\ &= \frac{|6.957 \times 10^8 - 4.625 \times 10^8|}{6.957 \times 10^8} \times 100 \end{aligned} \quad (110)$$

$$\text{\%Error} = 33.52\% \quad (111)$$

This result indicates that while the Lane-Emden equation provides a reasonable approximation, it cannot perfectly reproduce a realistic stellar model. This is primarily because the Sun has a radiative core, roughly modelled by $n = 3$, and a convective envelope, better described by $n = 1.5$ [1, §2.4]. A single polytropic index cannot capture this structural transition, limiting the accuracy of the model. Of course, there exist other factors such as nuclear energy production. Sedek et al. [6] extended this framework by developing composite polytropic models consisting of two and three distinct regions

corresponding to the Sun's convective, radiative, and nuclear zones. These models were calibrated against the Standard Solar Model of Bahcall, Serenelli, and Basu [7]. Their results showed that a three-zone composite polytrope reproduces the internal solar structure with reasonable accuracy, overcoming some of the limitations associated with describing the entire stellar body using a single polytropic index.

§3.3.2 Mass Comparison

In order to be able to numerically find M , we use the equation which describes the relation between mass M and radius r . However, since our analysis was done using the variable ξ and θ , we have to perform the following manipulation:

$$\frac{dM}{dr} = 4\pi r^2 \rho(r) \quad (112)$$

$$\int_0^R dM = 4\pi \rho_c \alpha^3 \int_0^{\xi_1} \theta^n \xi^2 d\xi \quad (113)$$

We can substitute the Lane-Emden equation into the integral, $\theta^n = -1/\xi^2 \cdot d/d\xi(\xi^2(d\theta/d\xi))$:

$$M = 4\pi \rho_c \alpha^3 \int_0^{\xi_1} -\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) \xi^2 d\xi \quad (114)$$

$$M = -4\pi \rho_c \alpha^3 \int_0^{\xi_1} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) d\xi \quad (115)$$

From the fundamental theorem of calculus and the boundary condition $d\theta/d\xi$ at $0 = 0$

$$M = -4\pi \rho_c \alpha^3 \xi_1^2 \frac{d\theta}{d\xi} \Big|_{\xi=\xi_1} \quad (116)$$

We can now use our numerical analysis and use the flag which finds the gradient at the surface in order to find the mass M . From Table 1, we know that for the polytropic index $n = 3$ the term $-\xi_1^2(d\theta/d\xi) |_{\xi=\xi_1} = 2.018$. We can thus compute M substituting the values of ρ_c and α which we had determined before and find it to be $M = 1.241 \times 10^{30}$ kg. We know from the standard value of solar mass $M_\odot$ to be $M_\odot = 1.989 \times 10^{30}$ kg, so the error we find is the following:

$$\begin{aligned} \% \text{ Error in Mass Determination} &= \left| \frac{M_\odot - M}{M_\odot} \right| \times 100 \\ &= \left| \frac{1.989 \times 10^{30} - 1.241 \times 10^{30}}{1.989 \times 10^{30}} \right| \times 100 \\ &= \left| \frac{0.748 \times 10^{30}}{1.989 \times 10^{30}} \right| \times 100 \\ \% \text{Error} &= 37.61\% \end{aligned} \quad (117)$$

The 37.61% error in predicted mass is comparable in magnitude to the 33.52% radius error found in §3.3.1. This consistency suggests both discrepancies share a common reason, the inability of a single polytropic index to capture the structural transition between the solar radiative core and convective envelope. A model that better reproduces one quantity would likely improve the other simultaneously, since both depend on the same underlying density profile.

§3.3.3 Density Profile

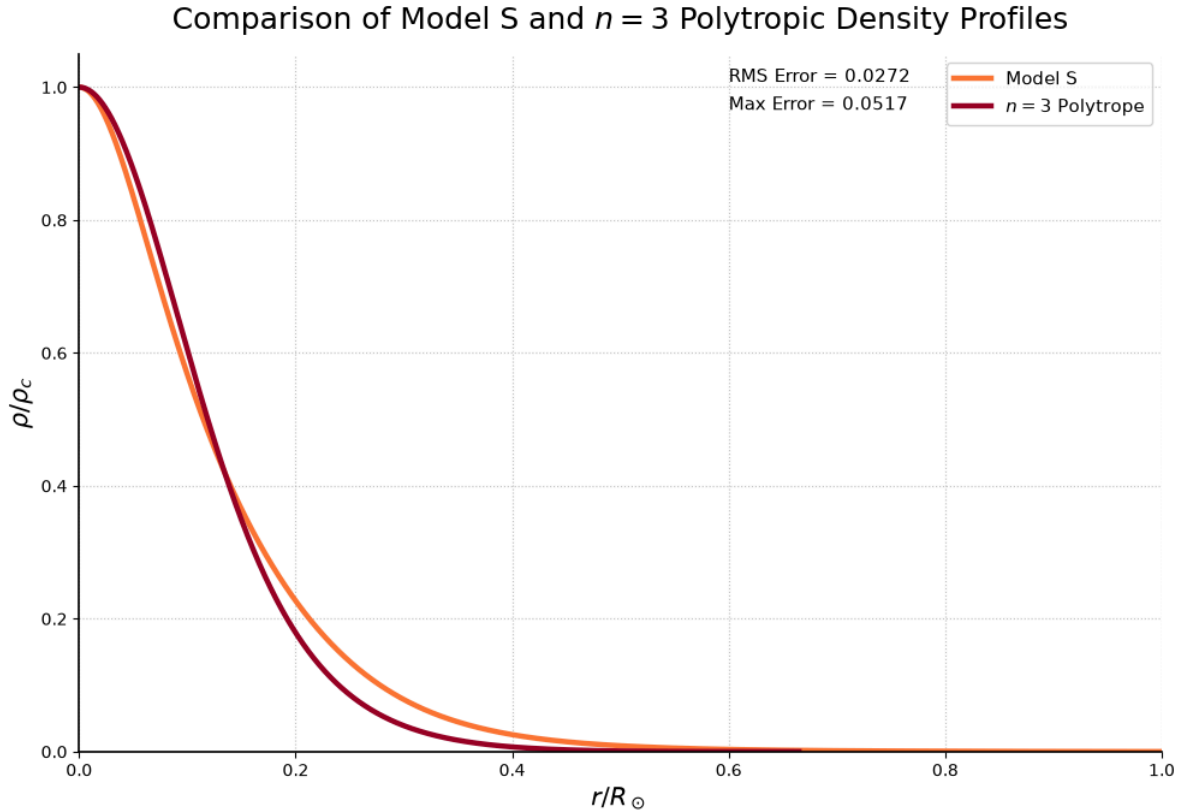
The Model S [5] has an online repository of the standard solar data. Using the repository and plotting it against the density profile of the Lane-Emden, we can graphically and numerically observe the accuracy of the Lane-Emden Equation to predict a density profile. The repository and the code which I used to plot it will be listed in the GitHub repository for ease of access. A more detailed explanation of the code is written in the code itself. However, I will explain the code generally here as well.

Using the *scipy* library in a similar fashion to §3.2. We can evaluate the function for $n = 3$ only. We can download the Model S data in a *.txt* file and use *np.loadtxt* in order to parse the text file. Storing the data in terms of arrays. Additionally, the Model S data is stored in terms of CGS units, so we convert to SI for familiarity and consistency. Moreover, the array is to be reversed as the Model S stores the data from surface to center. We can reproduce the calculations in §3.3.1 in order to evaluate the radius r and density ρ for each point from the Lane-Emden. There is a normalisation step, which constraints each of the graphical variables from 0 to 1. We have to perform an interpolation in order to compare the densities of the same radius. Which constructs a continuous density function of the Lane-Emden from which we evaluate the continuous density function at the points of the Model S only. We also find the RMS error and maximum error by computing it using:

$$\% \text{RMS Error} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\Delta \rho_i)^2} \times 100, \quad \% \text{Maximum Error} = \max_i |\Delta \rho_i| \times 100 \quad (118)$$

The maximum error is found by finding the index of the array which stores the maximum difference.

The following plot is obtained:



Some additional data which is printed in the terminal is shown in the table below:

Metric	Definition	Value	Interpretation
RMS error	$\sqrt{(\frac{1}{N}) \sum (\Delta\rho)^2}$	0.0272	2.72% Average Deviation
Maximum absolute error	$\max \Delta\rho $	0.0517	5.17% Largest Deviation
Radius of maximum discrepancy	$r/R_\odot$ at $\max \Delta\rho $	0.2277	Largest Deviation at $0.2277 R_\odot$

Table 2: Quantitative comparison between the Model S density profile and the $n = 3$ polytropic model.

From the data and comparison, we can observe that the Lane-Emden is able to predict the regions of the density with acceptable accuracy, even if the model is simplistic. It has the largest deviation at the region where the $r/R_\odot = 0.2277$. When comparing to the Model S, it is stated that in a similar region of $0.2277 R_\odot$, approximately $0.2 R_\odot$, the boundary of the nuclear burning core and the radiative core exists. This suggests that the discrepancy of Lane-Emden to predict this region stems from the singular polytropic nature of the curve, it is unable to model the behaviour of the boundary accurately.

Chapter 4. Conclusion

The goal of this paper/pedagogical review of the Lane-Emden was a way to understand with more depth and intuition of the Lane-Emden. The reason I find this to be an interesting way to get into stellar modelling as this is quite simplistic and can be understood without much mathematical background. However, it shows the nature of how stellar modelling is handled in a more scientific manner. Additionally, on a more personal note, this is done almost entirely by me and it was my first time trying to make a piece of educational content. It has also fueled my interest in the subject of stellar modelling. There will be changes in the document as needs be.

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